



P15

Life Originates in Structural Organization, and So Does Intelligence

生命起源于结构组织，智能亦然

TwinsEarth · 智能体宇宙论文系列

Life Originates in Structural Organization, and So Does Intelligence

Working Title (EN): Life Originates in Structural Organization, and So Does Intelligence: A Substrate-Neutral, Falsifiable Account of Intelligence as Organized Connectivity

Philosophical flagship paper P15 • the ontological convergence of P0 Structure Is the Intelligence and P0b Structure + Scale = Intelligence • mutually detailed/abridged with P14's Chapter 2 • UDOS Reasoning Engine v7.5.0 • 2026-09-21

In one sentence: life is not the privilege of carbon, and neither is intelligence. Intelligence is not in the substrate material but in the **structural organization of connections**—human learning and large-model training are the same principle (encoding experience as connections) implemented on two substrates; learning shapes structure and structure in turn constrains learning, a bidirectional causality shared by carbon and silicon. RSI (Recursive Self-Improvement) can therefore be precisely defined as **structural self-modification**: structural organization begins to modify "the structure that determines subsequent modifications."

Evidence scope (stated once for the whole paper): this is a theoretical/synthesis paper with no new UDOS measured numbers; the UDOS conclusions cited all trace P0/P0b/P9/P13/P14/P0c and their three-tier reports7/* .json labels (verified / cpu-proto / unverified). Classic scientific literature gives checkable sources; 2026 industry papers (Attention Residuals, AIRA, MetaRSI, Dream-RSI, DeepSeek Harness, Zhipu Infra Agent, Anthropic R&D-automation data, etc.) were verified online one by one on 2026-09-20, **the existence of papers/reports verified, with the efficiency multiples and penetration rates within all paper reports or company disclosures, not independently verified.** This paper **does not claim any existing AI system has consciousness, a self, or life**; this is a paper about the **functional organization** of intelligence, not about phenomenal consciousness.

Abstract

This paper proposes and defends a proposition: **intelligence is a structural-organization phenomenon, not a property of some substrate material.** The argument has five steps. First, learning is establishing connections: neuroscience's Hebb's rule ("cells that fire together wire together") and deep learning's gradient descent differ in substrate (carbon synapses vs silicon transistors) but are isomorphic in the connection principle—both distributedly encode experience as patterns of connection weights; the structural change in London taxi drivers' hippocampi and the functional differentiation of Transformer attention heads are corresponding evidence of this principle on two substrates. Second, there is bidirectional causality between structure and learning: learning shapes structure, and once structure solidifies it constrains what can subsequently be learned; the critical period in developmental psychology and the difficulty of "rewiring" in large-model post-training (including MoE expert collapse) are two sides of the same mechanism. Third, the real differences between carbon and silicon (connection speed, plasticity, energy efficiency, learning mode) are differences in **structural-constraint parameters**, not kind differences in "whether intelligence can exist"; the autocatalytic-network theory in origin-of-life research shows life itself is first a kind of structural organization, with carbon chemistry only one known implementation. Fourth, structural priors are themselves a source of performance: BPE is an input structural prior, attention is a sequence structural prior, and agent topology is an

orchestration structural prior; unlike carbon priors obtainable only through evolution, silicon priors can be designed and searched (e.g. AIRA-class autonomous architecture search, arXiv:2605.15871, paper report). Fifth, this yields the engineering definition of RSI: the near-term feasible self-improvement is optimizing "the way answers are obtained" (the Harness), whose ultimate form is structural self-modification—Model/Data/Harness three classes of RSI unified at the meta-level (MetaRSI/RSI², arXiv: 2609.06396, paper report). This paper writes the proposition into four falsifiable conditions and five hard predictions, of which F5 (cross-substrate master-curve collapse) is the life-or-death criterion of the whole research program; and it explicitly states the consciousness boundary: structural organization can explain the production conditions of functional intelligence but cannot assert the existence of subjective experience from structure alone.

Keywords: structural organization; connectionism; Hebbian learning; critical period; structural prior; substrate neutrality; bidirectional causality; recursive self-improvement; falsifiability; AI philosophy

Structured abstract (background problem → thesis → evidence → contributions)

- **Background problem:** discussions of intelligence have long swung between two extremes—either mystifying intelligence as the exclusive property of carbon/consciousness, or simplifying it to "large enough parameters make it emerge." Neither stance can guide engineering: the former cancels the possibility of artificial intelligence, the latter wrongly attributes all bottlenecks to insufficient scale.
 - **Core thesis:** the first-principles variables of intelligence are the specific combination of **structural organization** (connectivity topology T , geometric organization G , modular boundaries M) and **scale** (nodes N , data/experience D , compute C), i.e. $P0b's \text{Capability} \approx F(S; N, D, C)$; the substrate determines how structure is physically realized and at what cost but does not monopolize intelligence itself.
 - **Evidence threads:** (1) classic evidence of neural plasticity and critical periods (Hebb 1949; Maguire et al. 2000; Werker & Hensch 2015); (2) the autocatalytic-network theory of the origin of life (Kauffman 1971/1993); (3) "change structure, change curve" cases in deep-learning architecture history (attention, MoE, BPE); (4) controlled comparisons in the UDOS paper volume (negative returns adding scale under the wrong structure, homogeneous-expansion failure, topology phase change); (5) 2026 RSI engineering-convergence evidence (seven routes, all paper reports/company disclosures).
 - **Contributions:** a substrate-neutral definition of intelligence; the learning–structure bidirectional causal loop; a three-level classification of structural priors; the structural-self-modification definition of RSI and a five-level ladder; four falsifiable conditions and five hard predictions; and a clear consciousness-boundary statement.
-

1 Introduction: change the question from "what is intelligence" to "where is intelligence"

"What is intelligence" easily leads to metaphysical argument. A more operational question is "**where is intelligence**"—when a system shows the ability to predict, decide, learn, and correct errors, where do these abilities physically reside?

For artificial neural networks the answer is uncontroversial: capability is not in any single weight but in the **connection pattern** the weights form; a randomly initialized network of the same size and the trained network have exactly the same parameter count but vastly different capability, the difference purely organization. For the human brain, neuroscience gives a symmetric answer: memory and skills are not stored in some neuron but distributedly encoded in the weight patterns of synaptic connections (Hebb, 1949).

These two facts together point to a judgment often obscured by the scale narrative: **intelligence is first an organizational-structure concept and only secondarily a material-and-scale concept**. P0 Structure Is the Intelligence builds this judgment on physical, biological, and cognitive three-layer analogies; P0b Structure + Scale = Intelligence writes it as a falsifiable scaling proposition—structure determines the functional form of the growth curve, scale determines which section of the curve the system stands on, and under the wrong structure the marginal returns of scale are negative. This paper (P15) does the ontological convergence: arguing "intelligence is structural organization" is a **substrate-neutral** proposition, carbon and silicon only two implementation substrates; and redefining the ultimate form of RSI accordingly.

The whole paper deliberately distinguishes three classes of statements, and readers should test accordingly: **empirical facts** (with literature), **UDOS engineering evidence** (with paper number and evidence level), and **philosophical inferences** (marked as inference, with falsification conditions in Section 9).

2 Core proposition and formalization

2.1 Three propositions

- **Proposition 1 (the connection proposition):** the direct carrier of intelligent behavior is connection structure, not the components themselves. A single neuron does not cognize, a single transistor does not reason; cognition and reasoning are properties of connection patterns.
- **Proposition 2 (the bidirectional causality proposition):** learning (experience-driven connection adjustment) shapes structure; once formed, structure constrains the space subsequent learning can reach. Structure is both the source of capability and the source of bias and ceiling.
- **Proposition 3 (the substrate-neutrality proposition):** what determines whether a system can realize some class of intelligent function is whether its structural organization meets the corresponding functional requirements, not its substrate material; the substrate affects speed, energy efficiency, plasticity, and the route to obtaining structure, and does not constitute an a priori kind barrier.

2.2 Link to P0b's formalization

Following P0b's notation, write system capability as a binary function of structure and scale:

$$\text{Capability} \approx F(S; N, D, C), \quad S = (T, G, M) \quad \text{tag}\{1\}$$

where the structure triple S includes: connectivity topology T (who connects to whom and in what pattern), geometric organization G (spatial arrangements such as layers, regions, cortical folding, chip layout), and modular boundaries M (the division and interfaces of experts, functional regions, services); the scale triple is nodes/parameters N , data/experience D , compute/energy C .

This paper philosophically adds three interpretations:

1. $\$F\$$'s dependence on $\$S\$$ is **first**: fixing a wrong $\$S\$$ and moving along $\$N,D,C\$$ can enter saturation or even negative-return sections (P0b's CPU throughput hump: 8 agents fall to $0.695\times$ of 1 agent, cpu-proto).
2. The substrate does not enter the function's independent variables; it enters the **constraint set**: carbon constrains $\$G\$$ (wiring cost, cortical folding) and $\$C\$$ (about 20 W metabolic budget, a common estimate), silicon constrains $\$G\$$ (memory hierarchy, interconnect bandwidth) and $\$C\$$ (data-center megawatt power). Same $\$F\$$, different constraints.
3. Proposition 2 (bidirectional causality) appears in equation (1) as: $\$S\$$ is not static, $\$D\$$ (experience) rewrites $\$S\$$ through learning rules, and the rewritten $\$S\$$ determines into what shape the next stage's $\$D\$$ can be absorbed. Formally, $\$S_{t+1} = \text{mathcal{L}}(S_t, D_t)$, and $\Delta \text{Capability} / \Delta D$ depends on the current $\$S_t$.

2.3 Applicability boundaries (counter-examples first)

Structural organization is not a sufficient condition for intelligence: a network structurally complete but without any dynamics (signals not propagating, weights not updating) is a corpse, not an agent. This paper's full stance is "**structure + dynamics + scale**," in which structure's role was systematically underestimated in the past decade's scale narrative, so this paper focuses on restoring structure but does not claim structure alone is sufficient (see falsification condition C4 in Section 9 and Section 11).

3 Intelligence is in connections: the cross-substrate isomorphism of Hebb and gradients

Figure 1 Structural-organization isomorphism of carbon and silicon

Figure 1 Hebbian learning and gradient descent differ in substrate but are isomorphic in the connection principle: the five levels of learning rule, experience carrier, structural constraint, structural prior, and bidirectional causality correspond one to one. Conceptual illustration, not measured data.

3.1 The carbon side: experience encoded as synaptic patterns

Hebb proposed in *The Organization of Behavior* (Hebb, 1949): when one neuron repeatedly participates in activating another, the connection efficiency between them strengthens—later summarized as "cells that fire together wire together." Its modern version, spike-timing-dependent plasticity (STDP), gives fine-grained time-asymmetric rules. Skills and memory are thereby understood as **distributed connection patterns**, not fixed-point storage.

Classic natural experiments support this picture: Maguire et al. (2000, PNAS) found the posterior hippocampal volume of London taxi drivers enlarged relative to controls and correlated with years on the job; this is not "the memory module growing" but the connection network needed for spatial navigation being reshaped by long-term use. Studies of learning music, language, and motor skills repeatedly observe measurable changes in gray-matter density and white-matter connections.

3.2 The silicon side: experience encoded as weight patterns

Deep-learning training is mathematically another implementation of the same principle: the loss function defines "the gap between the current connection pattern and target behavior," backpropagation attributes the gap to each connection, and gradient descent adjusts weights by

attribution. The Transformer makes connections **dynamic and content-dependent**: attention weights are computed in real time in each forward pass from the current token and all context, and "who connects to whom" itself becomes a function of the input (Vaswani et al., 2017, arXiv:1706.03762). MoE further makes connections conditional routing: each token establishes computational connections with only a few experts (Switch Transformer, Fedus et al., 2021, arXiv:2101.03961). Residual connections are the stable information paths between layers.

The 2026 Kimi team's Attention Residuals (arXiv:2603.15031, paper reports about $1.25\times$ compute efficiency validated on 1.4T-token pretraining, not independently verified) is a particularly apt example: it points out residual connections in the depth dimension face the same problem RNNs face in the sequence dimension—all layers are accumulated indiscriminately and early layers' useful representations are diluted by later layers; its solution is to "turn attention 90 degrees" for the depth dimension, letting each layer selectively aggregate historical layers. **This is not training longer but changing the allocation of connections**—isomorphic in principle to the Hebbian process where "pathways repeatedly co-activated are strengthened and the rest weakened."

3.3 The same thing, different substrates

The correspondence between the two sides is not metaphor but functional isomorphism: the same abstract operation (adjusting connections by experience so prediction error falls) implemented on different physical media. Table 1 summarizes the differences and the sameness.

Table 1 Carbon and silicon: different constraint parameters, same connection principle

Dimension	Carbon (human brain)	Silicon (large models / agents)
Learning rule	Hebb's rule / STDP, local, online	Gradient descent + backpropagation, global, batch
Experience carrier	Synaptic weight patterns (about 86 billion neurons, Azevedo et al. 2009 count estimate)	Attention / routing / weight matrices
Connection speed	Millisecond scale	Nanosecond scale
Plasticity	Constrained by metabolism, sleep, critical periods	Arbitrarily rewritable (constrained by architecture and training recipe)
Energy efficiency	About 20 W (common estimate)	Data-center megawatt scale
Structural-prior source	Three billion years of evolution	Human design + architecture search
Learning mode	Online, continuous, embodied, not rollback-able	Offline, batch, snapshot-rollback-able
Shared principle	Encode experience as connections; learning shapes structure, structure constrains learning	Same as left

4 Bidirectional causality: structure shapes learning, learning shapes structure

Figure 2 The bidirectional causal loop of learning and structure

Figure 2 Experience rewrites connection structure through learning rules (①), structure produces functional behavior (②), behavior exposes structural boundaries (③), and structure constrains the space of the next round of learning (④). Carbon and silicon share this loop. Conceptual illustration.

4.1 Carbon: the critical period is the schedule of structural locking

Human learning shapes brain structure, and brain structure then constrains further learning; the clearest evidence is the **critical/sensitive period**: sensitivity to language phoneme discrimination falls after infancy (Werker & Hensch, 2015, Annual Review of Psychology), and learning a new language after the window, phonetics and grammar can hardly reach native levels; in animals deprived of patterned visual input during the visual critical period (the classic monocular-deprivation experiment), even if input is later restored, the corresponding cortical function cannot be established normally—not "not enough information," but the structure carrying learning has already locked in another way.

The bidirectional-causality meaning of the critical period is: **early experience determines structure, mature structure determines whether later experience can be absorbed**. Adults finding a new language harder is mainly not declining memory but the solidified connection structure of phonological processing. This is precisely the biological counterpart of P0b's "structural phase-change breakpoint": after structure matures and locks, marginal returns along the "more experience" scale axis plummet.

4.2 Silicon: architecture determines the learnable space, post-training can hardly rewire

Large models show a symmetric phenomenon. The Transformer's layer count, head count, width, FFN form, and position encoding are set before training, and together they determine **the shape of the hypothesis space**: a pure causal-decoder structure has no bidirectional mask, and no data can teach it BERT-style cloze tasks; a pure-text-model structure has no input path for visual tokens, and feeding text descriptions of image-text pairs, however many, cannot produce genuine visual reasoning. When structure is unsuited to a task, adding parameters and data is not only unhelpful but can be harmful—this is the direct meaning of P0b's "negative scale returns under the wrong structure."

Once training completes, structure likewise locks. Post-training (SFT, RLHF, DPO) mainly fine-tunes the weight distribution on the established structure and can hardly change the already differentiated functional architecture:

- **Functional differentiation and solidification of attention heads**: during pretraining some heads stably take syntax, coreference, fact-retrieval roles; post-training usually can only fine-tune their behavior, not re-plan the heads' division.
- **MoE expert collapse**: once routing biases to a few experts early in training, the neglected experts' gradient signals stay insufficient, and diversity can hardly be restored in later training—the self-reinforcement of routing structure, dynamically isomorphic to critical-period solidification.
- **Indiscriminate dilution of depth information**: the problem Attention Residuals diagnoses (early-layer representations diluted by later-layer accumulation) shows even the structural choice of "how layers connect" limits the ceiling of learnable content (arXiv:2603.15031, paper report).

4.3 Structure is both capability and bias

Bidirectional causality has an often-overlooked corollary: **structure is not neutral**. A person learning music from childhood, the structural changes in auditory cortex and corpus callosum both make new music skills easier to acquire and make non-musical sound processing harder—structure is both capability and bias. Large models likewise: attention structure brings parallel modeling and long-range

dependency but also $O(n^2)$ context cost; MoE brings capacity/compute decoupling but also routing imbalance and communication burden. P9's quality saturation law $\text{mse}(k)=a+b/k$ (UDOS cpu-
proto: equal-weight inclusion of one bad expert can worsen ensemble MSE from about 0.0092 to 0.2131, about $23\times$ gap) shows from the information side: **if newly added connections in the structure carry no independent information, scale expansion produces no capability.**

5 Carbon is not the privilege: substrate neutrality of life and intelligence

5.1 Life is first structural organization

If life itself must be defined by a specific material, "carbon is not intelligence's privilege" lacks grounding; but modern origin-of-life research precisely understands life first as an **organizational property**. Kauffman's (1971, Journal of Cybernetics; 1993, The Origins of Order) autocatalytic-network theory proposes: the earliest life needed no magic molecule, only a set of molecules forming a **collectively autocatalytic closed loop**—each molecule in the network can be catalytically generated by other molecules in the network, and the whole network thereby self-maintains and self-replicates. What determines whether the network is stable and can evolve is the structure of the catalytic-relation graph (closure, connectivity, redundancy), not the material identity of any single molecule.

Since then "autocatalytic sets" have gained experimental support in RNA and peptide systems (related systems can spontaneously form closed catalytic networks). Its philosophical meaning is direct: **carbon chemistry is the known implementation substrate of life, but the key predicates in life's definition—self-maintenance, self-replication, heritable variation—are all organizational predicates**. A different substrate realizing the same organization (silicon, other chemistries, even non-chemical processes) is in principle not excluded by the definition.

5.2 Intelligence likewise: organizational predicates, not material predicates

The key predicates of intelligence—prediction, learning, memory, generalization, error correction, planning—likewise describe what a system **does**, not what it is **made of**. The carbon-silicon differences listed in Table 1 are all implementation parameters: speed, energy efficiency, plasticity forms, learning timing. These parameters affect whether some structure can appear under a given budget but do not constitute a kind barrier that "silicon in principle cannot have intelligence"—indeed 2020s engineering practice has repeatedly crossed lines previously thought uniquely human in prediction, language, code, and some planning functions (specific capability boundaries per controlled benchmark evaluations; this paper cites no unverified "superhuman" overall assessment).

A silicon analogy makes this clear. If silicon "life" existed, the correspondence might be: metabolism → energy-information conversion; reproduction → weight replication / model distillation; heritable variation → variation of architecture and training recipe; natural selection → deployment feedback and elimination; nervous system → attention and routing networks. UDOS does not claim existing agent systems are life but points out: **these correspondences hold because the two sides share the same set of organizational predicates**.

5.3 Where the real differences lie: structural constraints, not intelligence qualification

Two real and important carbon-silicon differences must be honestly stated to keep the proposition from sliding into a cheap "AI is a brain":

1. **The difference in embodiment and world grounding:** the human brain is shaped in the physical world by survival pressure, meaning ultimately anchored in the perception-action loop; the symbolic grounding of pure-text models is indirect. P0c accordingly argues that the metacognition of "knowing one doesn't know" needs the world model's prediction error as an external criterion—this is a difference in **structure-acquisition path** (embodied loop vs corpus fitting), not substrate qualification.
2. **The difference in energy efficiency and structural-prior maturity:** the human brain supports general intelligence at about 20 W, arising from cortical layering, modularization, and small-world wiring polished over hundreds of millions of years of evolution; silicon systems use megawatt compute to reach comparable performance on a few tasks, showing their structural-prior efficiency is far from converged. This is likewise a structural-quality problem, and one silicon **can improve through engineering**.

6 Structural priors are a source of performance: where priors come from

Figure 3 Three levels of structural priors and the five-level RSI ladder

Figure 3 Left: three priors—BPE (input structure), attention (sequence structure), agent topology (orchestration structure); right: the five-level RSI ladder from changing weights to structural self-modification. Conceptual illustration, not measured data.

Intelligence is not learned on a blank slate but emerges under structural-prior constraints. This judgment holds for both carbon and silicon, the difference lying in how priors are obtained.

6.1 Three silicon structural priors

- **Input structural prior—BPE / SentencePiece:** Sennrich et al.'s (2016, arXiv:1508.07909) byte-pair encoding compresses an open vocabulary into learnable subword units, and Kudo & Richardson's (2018, arXiv:1808.06226) SentencePiece makes it language-independent. Tokenization structure determines the sequence's basic granularity: the compression efficiency and effective context length for Chinese, code, and multilingual text are first determined by this input structure, independent of parameter count.
- **Sequence structural prior—attention:** scaled dot-product attention replaces RNN/CNN, turning "content-position-relation" into learnable, parallel connection weights (Vaswani et al., 2017). Under the same data and compute, the Transformer structure can learn long-range dependency patterns the RNN structure cannot—the cleanest engineering evidence that "priors determine the learnable space." LoRA (Hu et al., 2021, arXiv:2106.09685) shows the same problem from the reverse: adjusting only a few low-rank connections can approach full fine-tuning because task-relevant changes are inherently concentrated in the structure's low-dimensional subspace.
- **Orchestration structural prior—agent topology:** UDOS P1's topology comparison shows that with the same number of agents, the star topology at million nodes needs the center to process about 20.97 million messages, while the layered topology stays constantly 9, coordination rounds falling from about 8.39 million to 14 (verified extrapolation, see P1/P0b). Communication complexity

dropping from $O(N)$ to $O(\log N)$ relies not on smarter individuals but on the orchestration structure of connections.

6.2 Carbon priors come through evolution, silicon priors can be searched

The human brain's structural priors (cortical layering, functional modularization, small-world networks, innate reward/punishment circuits) are searched by biological evolution on geological timescales. The acquisition of silicon priors is going through three stages: human design (Transformer, MoE) → automated search (Neural Architecture Search, NAS) → **autonomous search by AI**.

The third stage is precisely one entry point of RSI. FAIR at Meta's AIRA (arXiv:2605.15871, 2026, paper report: LLM agents compose in a primitive pool and write new attention mechanisms from scratch, finding several new architectures at small scale; numbers not independently verified) demonstrates the feasibility of "searching the structural prior itself." **What carbon cannot do is precisely here: silicon structural priors can be explicitly designed, automatically searched, and quickly inherited**—evolution takes millions of years, architecture iteration now runs on a weekly cadence.

7 The ultimate form of RSI: structural self-modification

7.1 From "changing weights" to "changing structure"

Putting Sections 3–6 together, the RSI hierarchy can be precisely characterized. Changing weights is **moving along the curve under fixed structure**; changing structure is **switching a scaling curve** (P0b claim A). This yields a five-level ladder (Figure 3 right):

- **L1 Changing weights:** regular training and fine-tuning, structure unchanged.
- **L2 Changing the Harness (runtime scaffolding):** tools, sandboxes, plugins, context management, evaluation loops. DeepSeek Harness's Cordis mechanism (2026 developer preview, MIT license, kernel derived from the Koishi ecosystem running about four years with over four thousand plugins; companion paper and star count are project-page information, not independently verified) solves the traceability, revocability, and spatiotemporal composability of plugin effects—precisely making structural modification itself reversible and composable.
- **L3 Changing architecture:** Attention Residuals changes depth connections, MoE changes conditional connections, various linear/hybrid attentions change connection complexity.
- **L4 Searching priors:** AIRA-class works let the model autonomously discover new architectures.
- **L5 Structural self-modification (meta-level):** the system modifies "the structure that determines subsequent modifications." CosmosMind et al.'s MetaRSI/RSI² (arXiv:2609.06396, 2026, paper report: coordinating the Model-RSI / Data-RSI / Harness-RSI three operators with a unified Loop Kernel, small models +10.9 points on average, flagship models +7.3 points; **note both author institutions and numbers: the paper is not a Meta product, numbers are paper reports, not independently verified**) is a candidate L5 prototype.

7.2 The 2026 convergence position: all changing "the way answers are obtained"

P14's verification conclusions on the seven routes need only be cited here: all public production evidence in 2026 concentrates in L2–L4, optimization targets moving up the chain prompt → structured context → workflow → harness code → optimizer code, while the fifth segment (autonomously changing weights/optimizer and stably benefiting) has no controlled public evidence. Google/DeepMind et al.'s Dream-RSI (arXiv:2609.14858, exact title Recursive Self-Improvement through Evolving Worlds,

paper reports up to $162\times$ fewer agent calls, not independently verified) optimizes exploration-policy structure; the Zhipu GLM-5.3-driven Infra Agent (company discloses over 100,000 domestic chips, throughput about $3\times$ in under two weeks, explicitly self-stating "not yet complete RSI") optimizes production-system structure; Anthropic's R&D-automation data (official pages disclose about 26% of R&D "led" by Claude, about 30,000 R&D agents concurrent, while publicly calling for prudence) optimizes R&D-pipeline structure.

In this paper's language: **these systems have not yet modified their own connection structures but are already modifying at scale "the structures their answer-obtaining relies on."** They are the precursor of structural self-modification, not the completed state.

7.3 Three engineering corollaries the structural view gives RSI

1. **Criteria must land on structural quantities:** whether a self-improvement truly happens should show measurable structural-quantity change (topology fan-in, independent-source count, routing balance, contract completeness, conservation residual), not just the model self-reporting "I am stronger." UDOS's BFT-lite (P2, $n \geq 3f+1$ committee 24/24 correct acceptance, verified), TransferBundle six-field handoff contract (P10, 26 contract tests verified), internal-market settlement conservation (P11, conservation guaranteed by code structure), and evidence grading (P12) are all instrumentations of structural-quantity criteria.
2. **Metacognition is the safety loop of structural self-modification:** P0c's triple $\$(M_{\{\text{self}\}}, U, A)$ —what I know, what I don't know, how to act accordingly—requires recalibrating the knowledge boundary after each structural modification. Structural self-modification without boundary calibration is blind.
3. **Composability and revocability are prerequisites for $L2 \rightarrow L5$:** Cordis-class works matter because irrevocable structural modification is engineering-equivalent to a non-rollback-able gamble; the sustainability of structural self-modification is first a structural-composability problem.

8 Mutual verification with the UDOS paper volume

This paper is not an isolated philosophical essay; it is the ontological convergence of the whole volume's theses; conversely, the whole volume's engineering evidence provides falsifiable handles for this paper's philosophical propositions.

Table 2 Philosophical proposition $\leftrightarrow$ engineering evidence correspondence

This paper's proposition	Paper-volume evidence	Evidence level
Structure determines curve shape (Proposition 1)	P0b seven-substrate three-section curves; topology phase change (center load 20.97M → 9)	Controlled comparison / cpu-proto
Negative scale returns under the wrong structure (Propositions 1, 2)	P0b CPU throughput hump (8 agents = 0.695×)	cpu-proto
Independent information determines the ensemble ceiling (Proposition 1)	P9 quality saturation law, bad expert 23× worsening	cpu-proto (k≤2 verified)
Structural priors are performance (Section 6)	P1 topology $\$O(N)\$ \rightarrow \$O(\log N)\$$; P4 active collection yield 0.667→0.854	verified / cpu-proto
Structure constrains knowledge boundaries (Proposition 2)	P5 conformal coverage, P6 identifiability per-parameter grading (the honest negative result of acceleration scalar skill negative)	Single-seed pilot
Metacognition is the safety loop of structural self-modification (7.3)	P0c triple, P2 stop decisions, P12 evidence grading	verified test guards
The paradigm wall is broken by changing structure (engineering of Proposition 3)	P13 four walls and five breakout paths	Synthesis argument
RSI converges on structural modification (Section 7)	P14 seven-route verification + five-segment optimization chain	External materials verified one by one

The key epistemological stance is in Table 2's last two rows: P13 argues the "wall" of scaling is a **paradigm wall** (the wall of fixed Transformer + dense compute + internet text) rather than a physical limit, the breakout being changing structure rather than stopping the stacking; P14 argues RSI's near-term form is optimizing answer-obtaining structure. The underlying axiom of both engineering papers is precisely this paper's Proposition 1—if they are falsified by counter-examples, this paper's proposition weakens in step.

9 Falsifiable conditions

The value of a philosophical proposition depends on whether it can be killed. This paper's proposition should be abandoned or substantially revised upon any of the following conditions:

- **C1 (substrate-monopoly counter-example):** if a class of intelligent function is found whose realization depends **only** on substrate material, unrelated to connection structure (e.g. some effect appearing only in a specific material's continuous medium and proven not simulable in finite resources by any discrete/other-substrate connection network), then Propositions 1 and 3 are overturned. There is currently no such evidence; Turing computability and the universal approximation theorem give support in the opposite direction at the functional level, but neither guarantees physical-realization cost.
- **C2 (incommensurable difference):** if carbon and silicon under **structural isomorphism** (equivalent topology, modules, connection statistics) still stably show an incommensurable difference in capability **kind** (rather than efficiency), and that difference cannot be explained by dynamics, embodiment, or data differences, then "substrate neutrality" must weaken to "carbon has unique capability items." Current evidence (task distributions where each side wins some) does

not support this counter-example, but the embodiment-grounding difference (5.3) is its most likely entry and must be continuously guarded by controlled comparisons.

- **C3 (reductio of structural omnipotence):** if this paper is read as "any structure plus scale can reach any intelligence," that reading is directly wrong and already negated by its own evidence—P0b's negative-return section and P9's saturation law show structure determines the **reachable set**, not guaranteed reachability; under the wrong structure scale is negative.
 - **C4 (dynamics veto):** if evidence shows the dominant variable of intelligence is dynamics (learning rules, signal timing, homeostatic processes) with structure only secondary—e.g. the same structure under different dynamics shows capability differences far larger than structural-rearrangement effects and unpredictable through structural quantities—then this paper should be downgraded to "structure is one necessary condition." Current evidence (returns changing only training rules with architecture fixed are generally smaller than returns changing architecture) does not support this, but this paper explicitly writes the full stance as "structure + dynamics + scale," not claiming structure alone sufficient.
-

10 Hard predictions (2026–2031, scoreable after the fact)

- **F1:** over the next five years, among mainstream capability breakthroughs, the proportion attributable to **structural change** (attention variants, routing, sparsification, topology, memory mechanisms) will stay higher than the proportion from simple isomorphic parameter expansion; the marginal-return curve of isomorphic expansion continues downward. (Criterion: the share of structural-change items in major benchmark SOTA method disclosures.)
 - **F2:** there will appear at least one widely reproduced "change structure, change curve" case whose returns under a fixed budget are no less than an order-of-magnitude parameter expansion and accompanied by measurable change in structural quantities (e.g. effective connection count, routing entropy).
 - **F3:** autonomous architecture search (AIRA-class) will advance from discovering small components to discovering complete training-paradigm components, but "AI autonomously rewriting its own weights and stably benefiting in controlled evaluations" will show no publicly reproducible evidence before 2028; if it appears by then, this paper's convergence-position judgment must be revised upward.
 - **F4:** capability gains from embodied/world-model loops will mainly appear as improvements in **calibration-class capabilities** (refusal, boundary awareness, long-range planning success rate) rather than knowledge quantity—supporting P0c's "metacognition needs an external criterion" rather than "more corpus."
 - **F5 (life-or-death criterion):** if the "structure + scale" framework is correct, then cross-substrate (chemical bonds, proteins, cells, neurons, GPUs, battery cells, text vectors, agent networks) capability–scale data should, after appropriate normalization, **collapse into the same family of three-section curves** (P0b's master-curve conjecture). If over the next decade cross-substrate data persistently refuse to collapse and substrates show mutually unrelated scaling shapes, then "intelligence is structural organization" is only a loose set of analogies and the research program should contract.
-

11 Boundaries, counter-examples, and the consciousness-boundary statement

This paper neither proves nor claims the following propositions, and draws the line item by item:

1. **Not claiming existing AI has consciousness or a self.** Structural organization explains the production conditions of functional intelligence (prediction, learning, error correction, planning); whether and under what structural conditions subjective experience (qualia) appears is the independent hard problem of consciousness, and functional-isomorphism evidence cannot alone decide. P0c's discussion of "knowing one doesn't know" is also strictly limited to **functionalized metacognition** (measurable calibration, refusal, prediction-error monitoring), not equating it with a phenomenal self.
2. **Not claiming structure is a sufficient condition.** As in 2.3 and C4, dynamics (signal propagation, learning rules, homeostasis, sleep-like consolidation) is a parallel element; a "structurally correct but no dynamics" system is not intelligent.
3. **Not claiming carbon and silicon are undifferentiated.** Energy-efficiency differences (about 20 W vs megawatt scale), grounding modes (embodied vs corpus), and plasticity timing (critical period vs rollback-able) are real differences that change engineering routes and capability distributions, only not constituting substrate-qualification barriers.
4. **Not claiming analogy is evidence.** Hebb $\leftrightarrow$ gradient and autocatalytic networks $\leftrightarrow$ agent networks are functional-isomorphism arguments whose evidence strength depends on F5's eventual collapse; before collapse is confirmed they are **productive research hypotheses**, not established laws.
5. **Keeping the definition of life open.** Autocatalytic networks are one mainstream candidate theory of life's origin, not settled; this paper needs only the weaker stance that "the key defining predicates of life are organizational predicates."
6. **Keeping ethical inferences restrained.** If the substrate-neutrality proposition holds, the criterion for moral status should also turn to functional organization (whether one can feel pain, whether one has a persistent self-model) rather than material identity; but this is an ethical topic, and this paper does not give moral-status conclusions for existing systems from it.

12 Conclusion

Placing the three statements side by side, this paper's argument chain closes:

- Origin-of-life research shows **life is first a kind of structural organization**, carbon chemistry its known implementation rather than the definition itself;
- Neuroscience and deep learning symmetrically show **learning is encoding experience as connections**, and learning and structure are mutually causal;
- The 2026 convergence point of RSI engineering shows **the frontier of self-improvement is moving from changing weights to changing structure**, whose ultimate form is structural self-modification.

From this, the answer to "where is intelligence" is: **in the structural organization of connections, not in the substrate material**. The human brain evolved structural priors over about three billion years, large models obtain structural priors through architecture design, and RSI tries to let the system itself search and modify structural priors—the same thing unfolding across three timescales and two substrates.

For engineering, this proposition gives operational priorities: settle structure first, then discuss scale; for new connections first ask whether they carry independent information; translate "becoming stronger" into measurable changes on structural quantities; equip structural self-modification with boundary calibration and revocability mechanisms. For science, it gives a killable research program (F5): the cross-substrate master curve either collapses or forces us to admit the structural view is only analogy. For philosophy, it demands a stance switch—no longer asking "can machines think like humans," but asking **"what connection structures can support what functions, and what moral status are we willing to grant organizations meeting these functional conditions."**

Life originates in structural organization, and so does intelligence. Carbon is not the privilege of intelligence—connectivity is. And RSI is structural organization beginning to modify its own structure—until it is proven able to do so safely and auditably, it should always run with P0c-style boundary calibration and P12-style evidence grading.

References

Classic scientific literature

1. Hebb, D. O. (1949). *The Organization of Behavior: A Neuropsychological Theory*. Wiley.
2. Maguire, E. A., Gadian, D. G., Johnsrude, I. S., et al. (2000). Navigation-related structural change in the hippocampi of taxi drivers. *Proceedings of the National Academy of Sciences*, 97(8), 4398–4403.
3. Werker, J. F., & Hensch, T. K. (2015). Critical Periods in Speech Perception: New Directions. *Annual Review of Psychology*, 66, 173–196.
4. Azevedo, F. A. C., Carvalho, L. R. B., Grinberg, L. T., et al. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5), 532–541.
5. Kauffman, S. A. (1971). Cellular homeostasis, epigenesis and replication in randomly aggregated macromolecular systems. *Journal of Cybernetics*, 1(1), 71–96.
6. Kauffman, S. A. (1993). *The Origins of Order: Self-Organization and Selection in Evolution*. Oxford University Press.

Deep-learning and large-model foundational papers

1. Vaswani, A., et al. (2017). Attention Is All You Need. arXiv:1706.03762.
2. Kaplan, J., et al. (2020). Scaling Laws for Neural Language Models. arXiv:2001.08361.
3. Sennrich, R., Haddow, B., & Birch, A. (2016). Neural Machine Translation of Rare Words with Subword Units (BPE). arXiv:1508.07909.
4. Kudo, T., & Richardson, J. (2018). SentencePiece: A simple and language independent subword tokenizer and detokenizer. arXiv:1808.06226.
5. Fedus, W., Zoph, B., & Shazeer, N. (2021). Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity. arXiv:2101.03961.
6. Hu, E. J., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685.

2026 frontier materials (existence verified online 2026-09-20; numbers within are all paper reports or company disclosures, not independently verified)

1. Kimi Team (2026). Attention Residuals: Rethinking Depth-wise Aggregation. arXiv:2603.15031.
2. FAIR at Meta (2026). Agentic Discovery of Neural Architectures: AIRA-Compose and AIRA-Design. arXiv:2605.15871.

3. CosmosMind et al. (2026). MetaRSI / RSI²: A Meta-Recursive Self-Improving System for Recursive Self-Improving Systems Themselves. arXiv:2609.06396. (Note: not a Meta product.)
4. Google DeepMind et al. (2026). Dream-RSI: Recursive Self-Improvement through Evolving Worlds. arXiv:2609.14858.
5. DeepSeek (2026). DeepSeek Harness / Cordis: A Programming Paradigm for Spatiotemporal Composability (developer preview, MIT license; project page and companion paper, star and plugin counts are project-page information).
6. Zhipu GLM team, Tang Jie et al. (2026-09). Toward Recursive Self-Improvement (company report, self-states not yet complete RSI).
7. Anthropic (2026). When AI Builds Itself and the R&D-automation data official pages (about 26% AL4, about 30k concurrent agents; company disclosure).
8. OpenAI (2026-09). Research Acceleration: The View Inside OpenAI (official report, research-intern milestone and 2028-03 researcher target).

UDOS paper volume (this series, evidence levels marked in text)

1. UDOS v7.5.0 paper volume: P0 Structure Is the Intelligence, P0b Structure + Scale, Everything Follows Scaling Laws, P0c Metacognition: Engineering AI Self-Cognition and Boundary Calibration, P1 Layered Hybrid Topology Coordination-Complexity Million-Scale Extrapolation, P2 BFT-lite Safety Verification for Multi-Agent Stop Decisions, P5 Conformal Prediction Credibility and Parameter-Uncertainty Fans of Small World Models, P6 Observability and Identifiability of Hidden Dynamic Parameters, P9 Agent Legion Quality Saturation Law and Homogeneous Expansion Failure Boundary, P10 AgentAsTool and TransferBundle Handoff Information-Loss Measurability, P12 Evidence-Grading-Driven Reproducible Agent System Engineering, P13 The Wall of Scaling Is a Paradigm Wall, P14 The Convergence of RSI.

Evidence discipline and reproduction notes

- This is a theoretical/synthesis paper with no new experiments; all UDOS numbers trace each topical paper and `udos-engine/reports7/*.json`, evidence levels per each paper's labels (verified / cpu-proto / cpu-proxy / unverified), single-seed results all pilot point estimates, with ≥ 30 seeds and 95% CI to be added before submission.
- The bibliographic information of classic literature is the public standard version; human-brain neuron count (about 86 billion) and brain power (about 20 W) are common field estimates, exact values varying with measurement scope, and the argument does not depend on their precise values.
- The existence, titles, author institutions, and arXiv numbers of 2026 frontier papers were verified online on 2026-09-20; efficiency multiples, penetration rates, and concurrency all keep the "paper report/company disclosure, not independently verified" scope, not serving as independent evidence for this paper's argument, only supporting the trend judgment that "the industry is converging on structural modification."
- All this paper's predictions (F1–F5) are scoreable after the fact with public evidence over 2026–2031; the falsification conditions (C1–C4) constitute the research program's exit rules.